

Range of Complex Engineering Activities

Complex activities mean engineering activities that have some or all of the following characteristics

Project: Bike Store Management System

Attributes	Bike Store Management System
A1 Range of resources	The project involves human collaboration, servers, databases, development tools, programming languages, and software resources. It uses frontend technologies, backend tools, MySQL database, and team effort to build and manage the web-based system.
A2 Level of interactions	As this is a group project, team members regularly coordinate with each other and consider the needs of store owners and potential users. The project requires interaction among developers, customers, managers, and system administrators to complete the system properly.
A3 Innovation	The project introduces a specialized bike store management solution that combines browsing, filtering, bike comparison, cart, recommendation, online test ride booking, manager dashboard, digital invoice generation, and sales reporting in one user-friendly platform for small retailers.
A4 Consequences to society/environment	The system has positive social and environmental impacts. It improves shopping convenience and price transparency, promotes helmets and safety gear, reduces paper use through digital invoices, lowers unnecessary travel, and helps reduce overstock through better inventory management.
A5 Familiarity	The activity includes some unfamiliar issues such as client-server architecture, secure database integration, customer profile management, product recommendation, invoice/report generation, and future extensions like online payment, mobile app, chatbot, and second-hand bike features.

Course Name: Junior Design | Course Code: CSE299 | Section: 12 | Group: 05